

PROFILE

PhD Candidate in Computer Science, expected December 2026, building reliable LLM and NLP systems across controllable generation, retrieval, agent orchestration, and evaluation. Published researcher with experience translating causal and robustness methods into reproducible Python, PyTorch, and FastAPI workflows.

EXPERIENCE

Graduate Research Assistant

University of Arkansas - Reliable and Controllable Language Models

- Design controllable-generation objectives combining representation intervention, contrastive learning, semantic preservation, and multi-metric evaluation.
- Co-developed CEBERT; achieved 85.9% accuracy on word-replaced data and 86.0% on misspelled data versus the strongest reported baselines at 70.3% and 73.7%.
- Developed a Minimum Description Length framework for adversarial detection without access to the original input; published in Entropy in 2024.
- Build evaluation pipelines spanning SBERT similarity, BERTScore, classifier metrics, LLM-as-a-judge workflows, experiment tracking, and failure analysis.

Aug 2021 - Present

Fayetteville, AR

SELECTED SYSTEMS

TraceLayer (Private case study) - Designed an evidence-centered resume verification workflow with deterministic identity gates, promotion-only evidence flow, bounded retrieval, and fixture-backed evaluation.

Market Brief Agents (Public repository) - Built a local-first research-to-video pipeline with LangGraph orchestration, typed manifests, RAG, audit bundles, validation gates, an operations dashboard, and deterministic fallbacks.

Hate Meme Detection (Public repository) - Refactored BERT, ViT, multimodal fusion, and Gemini baselines into a shared experiment package with consistent data, training, evaluation, prompting, and smoke-test interfaces.

PUBLICATIONS

- Tiwari, K., Zhang, L. (2024). [Implications of Minimum Description Length for Adversarial Attack in Natural Language Processing](#). Entropy, 26(5), 354.
- Tiwari, K., Yuan, S., Zhang, L. (2022). [Robust Hate Speech Detection via Mitigating Spurious Correlations](#). AACL-IJCNLP, 51-56.
- Tiwari, K., Zhang, L. Counterfactual Text Generation via Geometric Style Control in Large Language Models. Manuscript in preparation.

EDUCATION

Ph.D., Computer Science	M.S., Computer Science	B.Sc., CS & IT
University of Arkansas Expected Dec 2026	University of Arkansas Dec 2024	Tribhuvan University 2019

TECHNICAL SKILLS

Programming	Python, C++, SQL
LLM & NLP	PyTorch, Hugging Face Transformers, PEFT, TRL, Sentence-Transformers, BERT, SBERT
AI Systems	FastAPI, LangGraph, RAG, dense retrieval, Qdrant, Pydantic, Streamlit
Evaluation	BERTScore, embedding similarity, ranking metrics, LLM-as-a-judge, Weights & Biases
ML & Data	scikit-learn, XGBoost, Pandas, NumPy, spaCy, NLTK